Insert project title using sentence case and align centre



By

Name (Reg Number)

Project

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Commerce Honours Degree in Data Science and Informatics

Department of Information and Marketing Sciences

in the

Faculty of Business Sciences

at the

Midlands State University

Gweru/Harare

Insert Date, Month and year of submission

Supervisor: Insert the title and name of the supervisor

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DECLARATION

I, [**Name and surname**], hereby declare that I am the sole author of this project. The contents of this project represent my unaided work and that this project has not previously been published and/or submitted for academic examination towards any qualification.

Signature:..... Date: .....

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SUBMISSION APPROVAL

I hereby certify that, **Student (Registration Number R……..)** was under my supervision for the purpose of the Data Science and Informatics Project 2 in partial fulfilment of the requirements for the degree of Bachelor of Commerce Honours Degree in Data Science and Informatics. I also attest that **he/she** attended all revisions requirements to my satisfaction and participated in scheduled feedback sessions with me as the supervisor. I further confirm that the student demonstrated the functionality of the system to me. Therefore, the student has been granted permission to submit his/her project documentation for examination.

Name of the supervisor:

Supervisor's signature: ……..

Date : ……

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ABSTRACT

The student should write an informative abstract which summarises the study based on the following items:

- Background and problem
- Purpose of the project
- Methodology
- Results/findings
- Conclusions
- Key contributions
- Recommendations

A good abstract should: be precise; use no technical jargon; be brief **(250 - 300 words)** and specific; minimise the use of abbreviations; cite no references.

NOTE: The abstract should be written last in a single paragraph **WITHOUT** section headings. **In methodology section there is need to justify the model relevance. Model training and evaluation should be specified. In the results section visualization and interpretation should be specified. You need to specify how the model was deployed**

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ACKNOWLEDGMENT

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DEDICATION

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TABLE OF CONTENT

Note: Should be generated using MS word. Do not bold the items on the table of content.

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LIST OF FIGURES

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LIST OF APPENDICES

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ABBREVIATION/ACRONYMS

Abbreviation/Acronym	Full meaning	

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GLOSSARY

Term	Definition	

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CHAPTER 1: INTRODUCTION

1.1. Introduction [1-2 paragraphs]

Introduce the field or domain in which the project is situated. Explain the relevance and importance of the domain, highlighting any specific challenges, trends, or issues.

1.2. Background and context of the project [2 paragraphs]

The background and context section should provide a clear understanding of the project's purpose, justify the need for the work, and outline its importance within the specific domain. It sets the stage for the subsequent sections of the project report, helping readers grasp the motivation and significance of the work being undertaken.

1.2.1. Current state of knowledge [2-3 paragraphs]

Review relevant literature, studies, or prior work related to the project topic. Provide a brief overview of the existing knowledge or practices related to the problem or domain. Summarise relevant research, methodologies, or existing solutions that have been explored in the field.

1.2.2. Gap identification [single paragraph]

Identify any gaps or limitations in the current knowledge, methodologies, or existing solutions. Explain why existing approaches may be insufficient or why further investigation may be needed. The gap can be evidence based, knowledge, practical knowledge, methodological, empirical , theoretical, population.

1.3. Problem Statement [single paragraph]

Clearly state the problem or challenge that the project aims to address. Describe the specific issue or opportunity that motivated the project, emphasizing its significance and potential impact. The problem statement

1.4. Project aim

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The aim of a data science and informatics project can vary depending on the specific research question, problem statement, or objectives of the study. For example,

- Develop predictive model that utilize historical data to make accurate predictions or forecasts.
- Explore and visualize large datasets to gain a better understanding of the underlying patterns, trends, or anomalies.
- Apply machine learning algorithms or statistical techniques to identify patterns or anomalies in data.
- Develop optimization models or decision support systems that utilize data to optimize processes, allocate resources, or make optimal decisions.
- Build recommendation systems that provide personalized recommendations or suggestions to users based on their preferences, behavior, or historical data.
- The aim is to create a unified view of the data to enable efficient analysis, reporting, and decision-making.

Develop a machine learning algorithm to classify and detect fraudulent transactions in a financial dataset, improving fraud detection accuracy by at least 10%.

1.5. Project objectives

Objectives should be aligned to your problem, not just general aspirations. Outline **five (5)** specific objectives of the project. Specific objectives are the tangible and measurable outcomes that a data science and informatics project aims to achieve. These objectives should be specific, clear, and aligned with the overall aim of the project. The objectives should be function based and use action verbs.

1.6. Motivation of the project

Build on your inner calling: What problem do you feel compelled to solve? How will this project make a difference in your community or discipline? Reflect on your personal journey. What inspired you to pursue data science? Why this topic in particular? Use storytelling and life experience. Examples: family health data led to a medical AI app, or a passion for fairness led to a bias detection project.

1.7. Stakeholders

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Identify the key stakeholders involved in or affected by the project. This could include individuals, organizations, or communities that have a vested interest in the project's outcomes or can benefit from its findings.

1.8. Project scope

Define the scope and boundaries of the project. The scope defines the specific aspects or variables of the project that are intentionally included in the project.

1.9. Project assumptions

It is important to document project assumptions explicitly to ensure clarity and transparency throughout the project. Additionally, regularly reassessing and validating these assumptions as the project progresses can help mitigate risks and ensure the reliability of the project outcomes. The following are just examples of assumptions in data science and informatics projects.

a. Data quality assumption: It is assumed that the data collected for the project is accurate, reliable, and representative of the target population or system being studied.
b. Data availability assumption: It is assumed that the required data sources and datasets needed for the project are available and accessible within the specified timeline.
c. Assumption about data compatibility: It is assumed that the different data sources being used in the project are compatible in terms of format, structure, and quality, and can be integrated or harmonized effectively.
d. Assumption about the statistical distribution: It is assumed that the data follows a specific statistical distribution (e.g., normal distribution) or exhibits certain patterns that are relevant for the chosen analytical methods or models.
e. Assumption about relationships: It is assumed that there are meaningful relationships or correlations between the variables being analyzed, and that these relationships can be captured and modeled accurately.
f. Assumption about external factors: It is assumed that external factors (e.g., economic conditions, environmental changes) do not significantly influence the analyzed data or the project outcomes.
g. Assumption about generalizability: It is assumed that the findings, insights, or models derived from the project can be generalized to other similar contexts or populations.

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h. Assumption about model validity: It is assumed that the chosen machine learning or statistical models are valid and appropriate for the data characteristics and research objectives.
i. Assumption about resource availability: It is assumed that the necessary computational resources, software tools, and expertise required for the project are available or can be obtained within the project timeline and budget.
j. Assumption about stakeholder requirements: It is assumed that the defined project objectives, scope, and deliverables align with the needs and expectations of the project stakeholders.

1.10. Relevance and significance

Relevance and significance are two important aspects to consider in a data science and informatics project. Highlight the relevance and significance of the project in a broader context. Explain how the project contributes to advancing knowledge, solving real-world problems, or improving decision-making in the Zimbabwean society, industry, education, or governance.

1.10.1. Relevance

Relevance refers to the extent to which the project addresses a real-world problem or research question that is meaningful and applicable to a specific domain or context. A relevant project focuses on a topic or issue that is of interest or importance to stakeholders, such as businesses, organizations, or the scientific community. Relevance ensures that the project outcomes have practical value and can contribute to solving existing problems or improving decision-making processes.

1.10.2. Significance

Significance, on the other hand, relates to the potential impact or value of the project outcomes. A significant project generates results, insights, or solutions that have implications beyond the immediate project scope and can contribute to advancing knowledge, informing policies, or driving positive change. The significance of a project can be evaluated based on various criteria, such as the novelty of the research, the potential for practical application, or the potential to fill gaps in existing knowledge.

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1.11. Chapter summary

This section provides a concise **summary of the entire first chapter**, briefly recapping the main points discussed in each section. Provide a summary of the subsequent chapter in the project report.

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CHAPTER 2: LITERATURE REVIEW [12-15 pages]

2.1. Introduction to the Literature Review

The literature review section demonstrates your knowledge of the existing research landscape, establishes the rationale for your project, and positions your work within the broader academic or professional community. It provides a solid foundation for your methodology and helps identify opportunities for innovation and contribution. Explain the purpose and importance of conducting a literature review. Describe how it helps situate your project within the existing body of knowledge and provides a basis for your methodology and approach. It is important to note that references (in-text citations) are a must using APA 7th.

2.2. Research questions

Research questions are specific inquiries that the study seeks to answer. They guide the research process and help gather relevant data to address the problem. Clearly state the research questions that guide your literature review. These questions should be derived from your project's problem statement and objectives. Students should include a table which outlines the relationship between objectives and problem statement.

2.3. Search strategy and data sources

Describe the search strategy used to identify relevant literature. Explain the databases, journals, conferences, and other sources you utilized. Develop a comprehensive search strategy to identify relevant articles. Start by searching electronic databases, such as PubMed, Scopus, or Google Scholar, using a combination of keywords and controlled vocabulary terms. Also, consider searching other sources like conference proceedings, grey literature, and reference lists of relevant articles. Discuss any inclusion or exclusion criteria applied to select the most relevant studies. Screen the articles identified in the search based on their titles and abstracts, applying the inclusion criteria established earlier. Remove duplicate articles and exclude those that clearly do not meet your research question.

2.4. Data extraction

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Assess the full text of the remaining articles to determine if they meet the inclusion criteria. Read the articles thoroughly and note any reasons for exclusion. Mention how you selected which articles to include, how you removed duplicates, and how you filtered out irrelevant sources.
Create a data extraction form to systematically extract relevant information from the selected articles. This may include details on the study design, sample size, data collection methods, results, and key findings related to your research question. The articles should be a minimum of 20 which are refereed.

2.5. Quality assessment

Briefly describe how you ensured the sources are credible, peer-reviewed, or technically sound. This can be done using appropriate assessment tools or checklists specific to study designs (e.g., PRISMA checklist for randomized controlled trials).

2.6. Key concepts and themes

Identify and discuss the key concepts, theories, or themes that emerge from the literature you reviewed. This could include methodologies, algorithms, models, or frameworks commonly used in the domain.

2.7. Synthesis of findings

Summarize and synthesize the main findings and conclusions from the reviewed literature. Identify the commonalities, differences, or debates among the studies. Highlight any gaps or limitations in the existing research.

2.8. Methodological approaches

Discuss the various methodologies or approaches employed in the reviewed literature. Explain their strengths, weaknesses, and suitability for addressing the research questions or objectives of your project. Describe which methodologies are common (e.g., CRISP-DM, SEMMA, neural networks, statistical analysis). Link back to what you plan to adopt.

2.9. Relevant technologies or tools

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Identify any specific technologies, tools, or software commonly used in the domain. Discuss their applications, advantages, and limitations. Highlight any emerging trends or innovative approaches. What tools, programming languages, or libraries are often used? Are there standard platforms (e.g., Python, R, Tableau, Scikit-learn)?

2.10. Comparison with existing work

Compare and contrast your project's objectives with the existing research. Identify how your project builds upon or deviates from previous work. Explain any **novel contributions or unique aspects** of your project [how your project different is from what is already known or existing solutions?]. Discuss the limitations or gaps in the existing literature and methodologies.

How does your project relate to the reviewed work? What are you doing differently? What gap does your project fill?

2.11. Chapter summary

Summarise the key points of the chapter and briefly highlight the next chapter. Recap the chapter. Briefly explain what the literature says and how it has informed the direction of your project. Mention any models, algorithms, or ideas you will adopt in Chapter 3.

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CHAPTER 3: METHODOLOGY [8-10 pages]

3.1. Introduction

The Methodology section of a data science and informatics project outlines the approach, techniques, and tools used to collect and analyze the data and address the project objectives. It provides a clear and structured description of the steps taken to process the data, build models, perform analysis, and derive insights.

3.2. Research methodology

In the fields of data science and informatics, several methodologies are commonly used to guide the process of solving problems and extracting insights from data. These include but not limited to the following:

a. CRISP-DM (Cross-Industry Standard Process for Data Mining)

CRISP-DM is a widely used methodology for data mining projects that involves six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. It emphasizes understanding business objectives, iterative development, and integrating domain knowledge into the process.

b. KDD (Knowledge Discovery in Databases)

KDD is a comprehensive process that encompasses the entire knowledge discovery process, including data mining. It involves steps such as data cleaning, integration, selection, transformation, pattern discovery, evaluation, and knowledge presentation.

c. Agile Methodology

Agile methodologies, such as Scrum, are iterative and collaborative approaches commonly used in software development projects. They advocate for flexible planning, continuous feedback, and adaptive responses to changes. Agile methods can also be applied to data science projects, allowing for incremental development and quick iterations.

d. OSEMN

OSEMN (Obtain, Scrub, Explore, Model, iNterpret) is a data science methodology that provides a structured approach to analyzing data. It involves obtaining data, cleaning and preprocessing

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it, exploring, and visualizing the data, building models, and interpreting the results to gain insights.

e. SEMMA (Sample, Explore, Modify, Model, Assess)

SEMMA is a methodology developed by SAS Institute for data mining projects. It involves sampling the data, exploring, and visualizing it, modifying variables, building predictive models, and assessing the results.

f. Spiral Model

The Spiral model is an iterative and risk-driven approach commonly used in software development. It involves a series of iterations, each consisting of planning, risk analysis, development, and evaluation. The Spiral model can be adapted to data science projects to accommodate evolving requirements and mitigate risks.

NB: Kindly choose the model that applies to your research rather than select all.

3.3. Research design

Describe the overall research design or framework that guides your project. Explain whether it is exploratory, descriptive, predictive, or explanatory in nature. Justify the chosen approach based on the project objectives and the available data.

3.4. Data collection and preprocessing

The data collection and preprocessing section should provide a detailed account of the steps taken to acquire, clean, and prepare the data for analysis. It establishes the foundation for subsequent analysis and modeling, ensuring that the data is accurate, reliable, and appropriate for addressing the project's objectives.

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3.4.1. Data sources and acquisition

Describe the sources from which the data was obtained. This could include databases, APIs, surveys, existing datasets, or any other relevant sources. Explain the process of data acquisition, including any permissions or legal considerations.

NB: You are encouraged to use local datasets

3.4.2. Data variables and features

Identify and define the variables or features present in the dataset. Explain the meaning and significance of each variable, including units of measurement, data types, and any relevant metadata.

3.4.3. Data quality assessment

Discusses the importance of data quality assessment in ensuring the accuracy and reliability of the project results. Explain the steps taken to assess the quality of the data.

3.4.4. Data cleaning and preprocessing

Describe the procedures employed to clean and preprocess the data. This may include techniques such as removing duplicates, outliers, handling missing values, and correcting inconsistencies. Discuss any transformations, normalization, or scaling applied to the data.

3.4.5. Data integration and merging

If multiple datasets were used, explain the process of integrating and merging them. Discuss the criteria used for matching and combining the data and address any challenges or considerations specific to the integration process.

3.4.6. Feature engineering

Describe any feature engineering techniques applied to enhance the dataset's predictive power or extract relevant information. This may involve creating new features, aggregating data, or deriving meaningful variables from existing ones.

3.4.7. Dimensionality reduction

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Discuss any dimensionality reduction techniques used to reduce the dataset's complexity while preserving its essential information. This may involve methods such as principal component analysis (PCA) or feature selection algorithms.

3.4.8. Data Sampling

Explain any data sampling techniques employed, particularly if the dataset is large or imbalanced. Discuss the rationale behind the chosen sampling method and its potential impact on the analysis.

3.4.9. Data privacy and anonymization

Address any privacy concerns associated with the data. Explain the steps taken to anonymize or de-identify personally identifiable information (PII) to ensure compliance with privacy regulations or ethical guidelines.

3.4.10. Data Documentation

Highlight the importance of documenting the data collection and preprocessing steps. Discuss the creation of data dictionaries or data documentation files that provide a clear description of the variables, their definitions, and any transformations applied.

3.4.11. Data versioning and storage

Explain how the data was stored, organized, and versioned to ensure traceability and reproducibility. Discuss the use of data management systems or version control tools to maintain data integrity and facilitate collaboration.

3.5. Model selection

Discuss the selection of appropriate models or algorithms for the project. Explain the rationale behind choosing specific models based on the problem statement, data characteristics, and project objectives. Consider both traditional statistical models and machine learning algorithms.

3.6. Model training and evaluation

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Describe the process of training the selected models on the data. Explain the training and validation splits used, cross-validation techniques, or any other methods employed to assess model performance. Discuss the evaluation metrics used to measure the models' effectiveness.

3.7. Model tuning and optimization

Discuss any hyperparameter tuning or optimization techniques applied to improve model performance. Explain the methods used to fine-tune the models, such as grid search, random search, or Bayesian optimization. Justify the chosen approach and discuss the results.

3.8. Model validation and testing

Explain how the final models were validated and tested. Discuss the use of separate test datasets or holdout samples to assess the models' performance on unseen data. Present the evaluation metrics and discuss the generalizability of the models.

3.9. Statistical analysis

If relevant to your project, describe any statistical analysis techniques [descriptive, univariate, bivariate and multivariate Analysis] used to analyse data and test hypotheses, compare groups, or derive insights from the data. Explain the statistical tests employed, assumptions made, and the interpretation of the results and data visualisation techniques.

3.10. Software and tools

Identify and justify the software, programming languages, libraries, or tools used for implementing the methodology. This could include programming languages like Python or R, specific libraries or packages, or data science platforms.

3.11. Ethical Considerations

Address any ethical considerations associated with the methodology. Discuss privacy, confidentiality, bias, fairness, or other ethical issues that may arise during data analysis, modeling, or interpretation of results. Explain any steps taken to mitigate these concerns.

3.12. Chapter summary

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Summarize the key points discussed throughout the methodology chapter. Introduce the next chapter and restate the research objectives and explain how the chosen methodology addresses them.

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CHAPTER 4: RESULTS AND FINDINGS [12-15 pages]

4.1. Introduction

The Results and Findings section should present a clear and concise summary of the outcomes of the analysis, highlighting the key insights, relationships, and patterns observed in the data. It should provide a basis for interpreting the results, drawing conclusions, and making informed decisions based on the findings. This section of presents the outcomes of the analysis conducted and the insights gained from the data. It provides a comprehensive summary of the key results, patterns, relationships, or trends observed during the project.

4.2. Summary of key findings

Begin with a concise summary of the main findings and outcomes of the analysis. Highlight the most important insights or patterns discovered in the data that are relevant to the project objectives. Each sub-section focuses on a particular objective and presents the corresponding results.

4.2.1 Objective 1:

4.2.2 Objective 2:

4.2.3 Objective 3:

4.2.4 Objective 4:

4.2.5 Objective 5:

4.3 Descriptive statistics and visualizations

Present relevant descriptive statistics, visualizations, or charts that support the findings. Include summary statistics, histograms, scatter plots, bar charts, or any other appropriate visual representations of the data that help to communicate the key insights effectively. Ensure that the visual representations are clear, concise, and support the narrative. Provide captions or annotations to aid understanding. Use appropriate visualizations to effectively communicate the results and findings.

4.4 Model performance and evaluation

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Discuss the performance and evaluation metrics of the models developed. Present the accuracy, precision, recall, F1-score, or any other relevant metrics used to assess the models' effectiveness. Compare the performance of different models, if applicable.

4.5 Inferential Statistics

Describe any significant relationships, correlations, or patterns identified during the analysis. Explain the nature and strength of these relationships and discuss their implications in the context of the project objectives. Use visualizations or statistical measures to support your findings.

4.6 Statistical analysis results

If statistical analysis techniques were applied, summarize the results of the statistical tests conducted. Explain the outcomes of hypothesis testing, significance levels, and any relevant statistical measures that support or refute the project hypotheses.

4.7 Feature importance and insights

Discuss the importance and contribution of different features or variables in the models or analysis. Identify the key factors that strongly influence the outcomes or predictions. Explain the rationale behind these conclusions and their relevance to the project objectives.

4.8 Outliers and anomalies

Highlight any significant outliers or anomalies identified during the analysis. Discuss their potential impact on the results and the actions taken to handle or address these outliers. Explain the implications of these observations, if any.

4.9 Robustness and sensitivity analysis

If applicable, discuss the robustness of the findings and the sensitivity of the results to changes in the methodology or assumptions. Explore alternative scenarios or sensitivity analyses to understand the stability or variability of the conclusions.

4.10 Discussion and interpretation

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The Discussion and Interpretation section is where you delve into a deeper analysis of the results and findings presented in the previous section. It involves interpreting the implications of the results in the broader context of the project objectives, addressing any underlying assumptions, discussing potential explanations for the observed patterns, and providing insights into the practical significance of the findings.

Recap of key findings

Begin by summarizing the key findings and results presented in the Results and Findings section. Provide a concise overview of the most important insights derived from the data analysis.

Interpretation of Results

Interpret and explain the meaning and implications of the observed patterns, relationships, or trends. Discuss the underlying factors or mechanisms that may contribute to the findings. Consider both the statistical significance and the practical significance of the results.

Comparison with Prior Knowledge

Compare the findings with existing literature, prior studies, or established theories in the field. Discuss any consistencies, contradictions, or novel insights that emerge from the comparison. Identify areas of agreement or disagreement with previous work.

Validation and robustness

Discuss the robustness and reliability of the results. Address the limitations, uncertainties, or potential biases that may affect the interpretation of the findings. Consider alternative explanations or potential confounding factors that could impact the results.

4.11 Chapter Summary

Summarize the main points discussed in the Discussion and Interpretation section. Emphasize the key insights, implications, and recommendations resulting from the analysis. Provide a concise conclusion that ties together the findings and their significance to the project objectives.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS [4 -5 pages]

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5.1. Introduction

The Conclusion and Recommendations section provides a concise summary of the key findings, the implications of the results, and actionable recommendations based on the analysis. It serves as a final wrap-up of the project, highlighting the main takeaways and suggesting potential next steps.

5.2. Summary of findings

Begin by summarizing the main findings of the project by objectives. Briefly restate the key insights, patterns, relationships, or trends that emerged from the analysis.

5.3. Achievement of objectives and conclusions

Evaluate the extent to which the project objectives were achieved based on the findings. Discuss the relevance and significance of the results in addressing the research question or problem statement.

5.4. Implications and impact

Discuss the broader implications and impact of the findings.

5.4.1. Implications to theory

Explain how the results contribute to the existing knowledge in the field, address gaps, or provide new insights.

5.4.2. Implications to practice

Highlight the practical significance of the findings and their potential impact on decision-making or interventions.

5.4.3. Implications to education 5.0

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This subsection specifically examines the implications of the research findings for education, particularly in the context of Education 5.0. It discusses how the results can inform and shape educational practices and approaches.

5.5. Limitations and caveats

Acknowledge the limitations and caveats of the project. Discuss any constraints, uncertainties, or potential sources of bias that may have influenced the results. This could include factors such as data availability, time constraints, budget limitations, or technical considerations.

5.6. Recommendations

Provide actionable recommendations based on the findings. Discuss specific actions, strategies, or interventions that can be implemented to leverage the insights gained from the analysis. Provide clear and practical recommendations that are aligned with the project objectives on future work or improvements.

5.7. Future research directions

Identify potential avenues for future research or investigation based on the findings. Highlight areas that require further exploration, refinement, or validation. Suggest additional data sources, methodologies, or models that can enhance the understanding of the research question or problem. Identify any unanswered questions or areas that require more in-depth exploration. Provide recommendations for refining the methodology, expanding the dataset, or exploring related research topics.

5.8. Conclusion

Conclude the project by summarizing the main points discussed in the conclusion and recommendations section. Emphasize the value and contribution of the project to the field. Provide a final thought or reflection on the significance of the work conducted.

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References

List of cited references, papers, and resources used in the project using APA reference guidelines.

List of Appendices

Appendix A: AI declaration form

Appendix B: Data Extraction Form

Appendix C: Technical details of the project

Appendix D: Code snippets or algorithms

Appendix E: Data dictionaries

Appendix F: Turnitin Report [Similarity index of 20% or below]

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AI ACKNOWLEDGMENT GUIDE

Midlands State University, Department of Information and Marketing Sciences' approach to the use of AI technologies requires you to **openly and transparently acknowledge how and why you have used it. Acknowledgment** simply means describing how **you have used Generative AI in the process of creating a work for submission.** While you may not have used the content directly as a source of information in a quotation or citation, to maintain good academic practice and the fairness of assessment, **you must acknowledge this contribution.**

The emphasis is on demonstrating critical use of the tools, questioning their value, accuracy, and appropriateness for the task. Articulating how and where AI has informed or supported your work will allow you to demonstrate the development of your own learning while avoiding academic misconduct. There are many ways that you may have used AI tools in the preparation of an assessment. These could be tools built into existing applications or stand alone. While you may not have used content directly as a source of information in a quotation or citation, you may have used it as a digital assistant in the creation of your assessed work. **To maintain good academic practice and the fairness of assessment you must acknowledge this contribution.**

How do you acknowledge the contribution of AI?

It is good academic practice to provide a summary **acknowledging how, why, and when you have used AI technologies**. Describing not only what you did but why will help you be more critical in your selection and use of these tools, consider the role of AI in the learning process, and understand its impact on your work. This will be done using AI Declaration Form or An Appendix with the following.

- a list of the AI tools used and why you chose them
- a brief explanation of what you used the tool for (for example, to gather information, to aid your understanding of a concept, topic or course materials, to generate new ideas or plan your writing, to check facts, or to analyses data)
- example prompts you might want to highlight, including annotations or a comment on why they were useful.

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AI DECLARATION

I **(Full Names and Student Registration number)** understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by the department as academic misconduct. I declare that while conducting the research towards my degree or in the preparation of this thesis/dissertation/research report /project/assignment (select one by marking with an X).

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☐
2. Sourcing Related Work (summarising, identifying sources) ☐
3. Methods and Experiment Design (experiment setup, model tuning ☐
4. Data Analysis (presentation, coding, interpretation) ☐
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☐
7. Presentation (rendering graphics, formatting) ☐
8. Editing (grammar, readability) ☐
9. Writing (text generation, document structuring) ☐
10. Citation Formatting (structuring, organising) ☐

If other uses were involved, please specify below

Generative AI tool used (list all)	**Used for?**

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my thesis/dissertation/research report /project/assignment, I confirm that details of this use are clearly outlined in the relevant methodology chapters/or appendix to my thesis/dissertation/research report /project/assignment report.

Candidate signature:

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Supervisors and supervisees should pay attention to the following formatting:

Font type: Times New Roman throughout the document

Font size:

- Black-Bold and 14-point for Heading 1 (Chapter headings e.g. Chapter 1, Chapter 2, etc and headings on preliminary pages, e.g. Abstract, Declaration, etc.) and Heading 2 (e.g. 1.1 Introduction;)
- Black-Bold and 12-point for Heading 3 (1.4.1 Sub-research question) and heading 4. **NOTE:** The section headings should not go beyond heading 4.

Section headings: Sections headings should be aligned left, bold and numbered. Make sure you are consistent in your numbering of headings and subheadings.

Paragraphs alignment: align all paragraphs to justify

Saving and submitting the document: Save the document with the following format; name_surname_registration number,

NOTE: The document should be in .doc format and PDF.

Spacing: set spacing to 1.5 between the Heading/subheading and the first line of the paragraph; 1.5 line spacing (within the paragraph); 'cursor' spacing between paragraphs.

Margins: Top-2.54 cm, Bottom-2.54 cm, Right-2.54 cm, Left-2.54 cm, Gutter -0 cm, Gutter position-Left.

Page numbering: No numbering of the cover page; preliminary pages should be numbered in Roman numerals, the bottom of page plain number 2 format. Check the numbering done in this document.

Tables and Figures: Font size, numbering and labelling

Tables must be numbered consecutively using Arabic numbers throughout the dissertation, as should figures. Each table or Figure in a dissertation must have a caption that tells concisely what it contains. The caption of a Table must be placed **above** while the caption of the Figure must be placed **below**. Tables and Figures should include a chapter designation as well as a Table and Figure number (Table 1.1, 1.2, 1.3; 2.1, 2.2, 2.3, 3.1, 3.2) both in the text

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and in the caption. Font size for the Table caption and text inside the table should be **10** while spacing should be **1.15.** See examples below:

Table 1.1: F1-Score

	0.85
	0.72

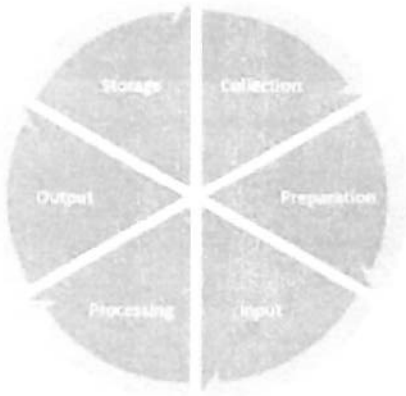


Figure 3.1: Data processing cycle

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